

Yahia Abubashem

Lead Data Scientist | PhD in Economics

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Professional Summary

Lead Data Scientist specializing in AI innovation and data-driven transformation across healthcare and insurance ecosystems. With over 8 years of experience, I design and deploy machine learning systems, generative AI workflows, and predictive models that directly enhance clinical outcomes, population health, and operational performance. Expert in causal inference, LLM integration, retrieval-augmented generation (RAG), and agentic workflows, bringing advanced AI techniques into regulated, high-stakes healthcare environments. Proven ability to align AI strategy with business and compliance goals, delivering scalable, explainable, and ethical models under MLOps best practices using AWS, Databricks, and MLflow. Known for leading cross-functional teams, mentoring emerging scientists, and translating complex analytics into actionable intelligence for executives, clinicians, and policy leaders.

Technical Skills

Programming & Data Tools: Python (Pandas, NumPy, Scikit-learn, PySpark), SQL, R, Jupyter, Git, Docker, Kubernetes, Airflow, Databricks, MLflow

Machine Learning & AI: TensorFlow, PyTorch, XGBoost, LightGBM, LLMs, RAG, Agentic Workflows, SHAP, LIME

NLP & GenAI: spaCy, NLTK, Transformers, BERT, GPT-3/4 APIs, LangChain, Vector DBs

Computer Vision: OpenCV, Detectron2, CNNs, ResNet, Vision Transformers

Cloud & Infrastructure: AWS(SageMaker, Bedrock, Lambda, Glue, Redshift), Azure ML, GCP AI Platform, Snowflake, Data Lakes, CI/CD Pipelines

BI & Monitoring: Tableau, Power BI, Looker, Prometheus, Grafana, ELK Stack

Testing & Deployment: FastAPI, Flask, REST APIs, Postman, A/B Testing, Feature Flags

Agile Tools: Jira, Confluence, Git, Trello, Slack

Soft Skills

Cross-Functional Leadership, Communication & Data Storytelling, Mentorship, Collaboration & Stakeholder Engagement, Innovation Mindset, Results Orientation

Professional Experience

Lead Data Scientist

CVS Health

07/2020 – Present | Greater Chicago Area

- Spearheaded development of AI/ML models improving pharmacy benefit management and clinical decision support, impacting millions of members nationwide.
- Designed and deployed predictive models for drug utilization, adherence forecasting, and care cost optimization, reducing unnecessary spend by ~15%.
- Integrated generative AI and RAG pipelines for claims summarization and document classification, cutting manual review time by 40%.
- Applied causal inference, propensity modeling, and time-series forecasting to measure intervention outcomes and inform benefit design.
- Partnered with clinicians, actuaries, and engineers to embed AI-driven insights into patient engagement and formulary strategy.
- Standardized MLOps practices (using AWS, Databricks, MLflow, and CI/CD pipelines) to ensure model reproducibility and regulatory compliance.
- Mentored a team of data scientists, establishing best practices for data governance, model explainability, and ethical AI in healthcare.

Senior Data Scientist

AIM Specialty Health

06/2019 – 07/2020 | Greater Chicago Area

- Managed cross-functional teams delivering ML-driven diagnostics optimization for oncology and radiology service lines.
- Built risk stratification and utilization forecasting models using Python, scikit-learn, and SQL, improving prior authorization accuracy by 20%.
- Established an analytics experimentation framework to accelerate A/B testing and improve evidence-based decision making.
- Designed automated Power BI dashboards for patient outcomes, provider compliance, and cost of care monitoring.
- Collaborated with engineering teams to deploy models through scalable data pipelines and API integrations aligned with enterprise architecture.
- Delivered presentations translating ML insights into clinical and business actions, driving measurable ROI for care management programs.

Senior Data Scientist

World Bank Group

03/2015 – 07/2019 | Washington, D.C

- Led data science initiatives analyzing healthcare access, financial inclusion, and population health trends across developing economies.

- Built forecasting models, simulation frameworks, and econometric pipelines using Python, R, and SQL to evaluate national health reforms.
- Delivered high-impact insights to policymakers, influencing resource allocation and health program design in 10+ countries.
- Implemented reproducible analytics workflows and trained local analysts in AI-assisted econometrics and datadriven policy design.

Research Consultant

American University

08/2013 – 06/2018 | Washington, D.C

- Taught graduate-level courses in statistical modeling, econometrics, and applied data science.
- Developed reproducible frameworks for experimental and quasi-experimental health policy research.
- Published and supported academic work bridging economic modeling with applied machine learning for healthcare outcomes.

Education

American University
Doctor of Philosophy - Economics

2014 – 2019 | Washington, D.C

American University
Master of Arts - Economics

2013 – 2014 | Washington, D.C

Roanoke College
Bachelor of Arts - Economics

2008 – 2012 | Salem, Virginia